

# TrustShield AI

AI-powered weekly parametric income protection for gig delivery workers

- Coverage scope: loss of income only due to weather, pollution, and curfew disruptions.
- Excludes health, life, accident, and vehicle repair by design.
- Built for adversarial environments with explainable AI and fraud controls.

# Problem and Persona

Delivery workers lose 20-30% income during uncontrollable external disruptions

- Target persona: platform-based delivery worker (food, grocery, e-commerce).
- Workers need fast weekly protection aligned to weekly earnings cycles.
- Current safety nets are weak, manual, and vulnerable to false or delayed claims.

# Our Solution

Weekly micro-insurance with AI decisioning and automated claim routing

- Role-based surfaces: Worker, Insurer/Provider, Platform Ops, Admin.
- Provider-linked data isolation ensures insurer portfolio boundaries.
- Real-time disruption evidence drives claim simulation and approvals.

# How It Works

End-to-end flow from signup to payout in one trust pipeline

- Worker signs up, selects provider, verifies email, then awaits admin activation.
- Worker buys weekly policy and submits telemetry-backed claim trigger.
- AI + fraud + anomaly + graph signals produce APPROVED, NEEDS\_REVIEW, or REJECTED decisions.

# AI and Fraud Architecture

Clear, explainable logic over black-box complexity

- Composite risk = location + device + behavior + network + cluster risk.
- Fraud score stack includes GPS inconsistency, weather mismatch, repeat claims, and linked accounts.
- Graph intelligence identifies synchronized claim pressure and ring-like behavior.

# Phase 3: Advanced Fraud Detection

Delivery-focused practical scenarios implemented

- GPS spoof pressure: unrealistic speed jumps and confidence mismatch flags.
- Fake weather claim check: trigger data cross-validated with disruption signals.
- Mass pattern detection: synchronized location claims increase review pressure.

# Phase 3: Instant Payout Simulation

Fast payout experience with gateway-grade transaction trace

- Approved claims now carry payout status, gateway, transaction ID, timestamp, and processing seconds.
- Gateway modes supported: RAZORPAY\_TEST, STRIPE\_SANDBOX, and UPI simulator fallback.
- UI highlights under-30-second payout SLA for judge-visible impact.

# Intelligent Dashboard

Worker trust view + insurer intelligence layer

- Worker dashboard: earnings protected, claims history, latest payout trace, active coverage signals.
- Insurer dashboard: loss ratio, next-week disruption risk band, fraud pressure, payout SLA.
- Ops and Admin dashboards provide queue controls, governance actions, and review workflows.

# Weekly Pricing Model

Financial logic built for gig-worker weekly cash flow

- Premium calculated weekly with dynamic risk modifiers and provider product controls.
- Loss ratio tracking improves underwriting and reserve posture decisions.
- Provider liquidity and payout ledger keep portfolio sustainability visible.

# Business Viability

B2B2C-ready architecture with clear monetization levers

- Providers publish policy products and underwrite linked worker cohorts.
- Platform earns distribution and risk-management value through trust automation.
- Data isolation and governance controls improve enterprise adoption confidence.

# Scalability and Defensibility

Built to scale across cities, providers, and disruption patterns

- Queue-based processing for ingestion, model jobs, replay, and retries.
- Feature snapshots and model artifacts support continuous learning loops.
- Modular API design supports future real gateway and platform integrations.

# Why TrustShield AI Wins

Fast, explainable, fraud-aware income protection for India's gig economy

- Strong phase coverage: ideation, automation, and scale optimization complete.
- Clear demo story: disruption trigger -> AI decision -> instant payout simulation.
- Balanced approach: worker trust, insurer control, and platform resilience in one stack.